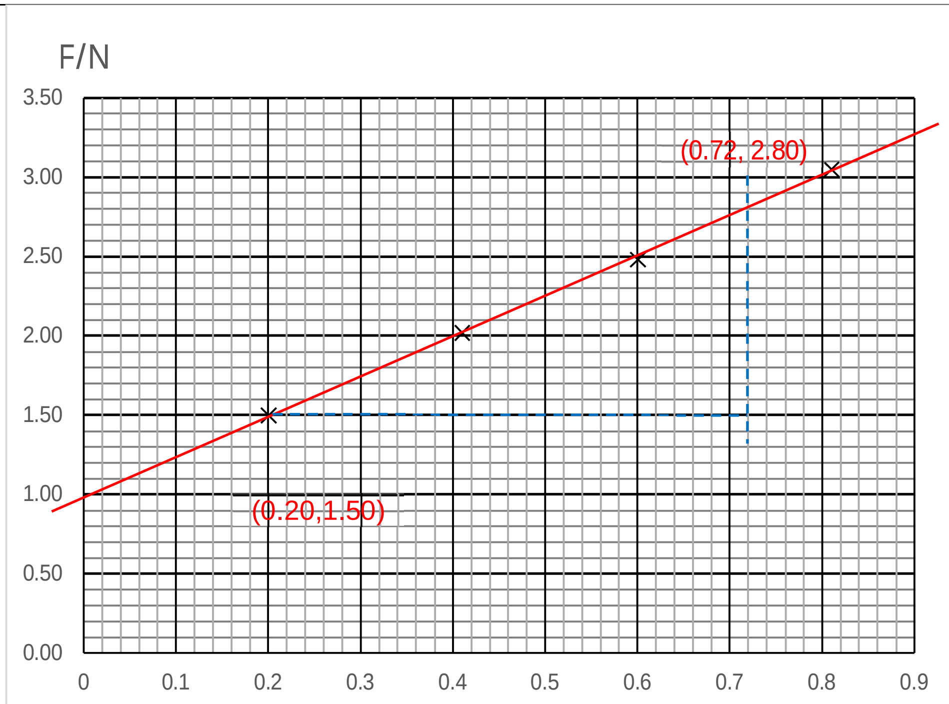
Qns	Answer	Marks
4(a)	Magnetic flux density *B* is the force acting per unit current per unit length on a wire carrying a current that is normal to the magnetic field.	**B1**
b(i)	Both arrows point vertically downwards (in diagram). The current flows perpendicular to the magnetic flux density at all parts of the coil. Hence the current carrying conductor will experience a magnetic force. The direction of the magnetic force can be deduced by Fleming's Left Hand Rule.	**B1** **B1** **B1**

(b)(ii)



B1 correct plots

B1 Best fit line

$$\text{Gradient} = \frac{2.80 - 1.50}{0.72 - 0.20}$$
$$= \frac{1.30}{0.52}$$
$$= 2.5$$

Force per unit current = F/L = 2.5 N A^{-1}

B1 use of gradient rather than plots, within range

From the graph, zero error of the balance =1.00 N

OR

Use point (0.20,1.50) and gradient =2.50 to find the y-intercept:
1.50 = (0.20)(2.80) + C
C = 0.94
Zero error = C = 0.94 N

B1

(iii)

$$F = BIL \Rightarrow \frac{F}{I} = BL$$

From (b)(i), $\frac{F}{I} = 2.5 \text{ N A}^{-1}$,

$$2.5 = B\left(50 \times 2\pi\left(0.025\right)\right)$$

M1

$$B = 0.637 \text{ T}$$

A1

Qns	Answer	Marks
5(a)	The gravitational field strength at a point is the **gravitational force** (of	